# Lekhraj Mahajan

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## Education

**Silver Oak University , Ahemdabad** 2023 - 2026
*Bachelor of Technology – Computer Engineering , CGPA - 8.60*

**Government Polytechnic , Bhuj** 2020 - 2023
*Diploma – Computer Engineering , CGPA - 7.75*

## Work Experience

**MERN Stack Developer - Tech Creature Solutions** May 2026 – Present
*Full-stack work on The Spot Exam, an enterprise examination SaaS - Ongoing Project* Gandhinagar, Gujarat

- Developed the Node.js / Express / TypeScript backend with 80+ modular domain modules (controller–service–repository pattern) on MongoDB/Mongoose, with Zod request validation and centralized error handling.
- Implemented JWT authentication and a granular RBAC / permission system across 10+ roles (Master Admin, Company Admin, Exam Manager, Center Manager, Observer, Paper Setter, Biometric Verifier, Candidate), with audit logging, rate limiting and Helmet-based security hardening.
- Built real-time proctoring and live monitoring with Socket.IO (Redis adapter for multi-instance scaling); used BullMQ job queues, node-cron schedulers and Redis caching for background processing and performance.
- Built the React + TypeScript admin portal with role-specific layouts, Zustand state, React Hook Form + Zod, route-level code splitting (lazy loading) and PWA support.
- Developed the candidate exam client (React, Vite) with in-browser face verification (face-api.js) and violation detection – tab switch, focus loss, copy/paste, no-face and multiple-face events.

## Projects

**The Spot Exam** | *Node.js, Express, TypeScript, MongoDB, Redis, BullMQ, Socket.IO, React, Zustand, Tailwind CSS*
*Multi-tenant online examination, proctoring and exam-center management platform* Live: thespotexam

- Designed a multi-tenant SaaS architecture for coaching companies and government-style exam bodies: three apps (admin portal, candidate exam client, REST API) with subscription plans, billing and invoicing.
- Built the exam lifecycle end to end: question bank, paper setting, shifts, seat allocation with QR codes, bulk candidate import (Excel/CSV), live answer capture and secure submission.
- Added integrity layer: face verification, per-candidate trust score, geo-monitoring of exam centers, live candidate tracking and violation logs for observers.
- Integrated Razorpay payments, Nodemailer, Firebase push notifications and Cloudinary; added support tickets, API metrics, Winston logging and PM2 deployment.

**AURA Headphones** | *React, Node.js, Express, MongoDB, Redis, Firebase Auth, GSAP, Cloudinary*
*Full-stack premium audio e-commerce platform with AI product recommendations* Live: auraheadphones

- Built a full-stack e-commerce flow (catalog, product details, cart, checkout, order tracking, user profile) with a React frontend on Vercel and Node/Express REST API on Render.
- Implemented Firebase Authentication (email/password and Google sign-in) with server-side ID-token verification, JWT-protected routes and a separate admin role middleware.
- Improved performance with Redis caching (1-hour TTL) for product APIs, express-rate-limit, Cloudinary image optimization and lazy-loaded routes; added GSAP/Lenis scroll animations.
- Built an admin dashboard to manage products, orders, users and discount coupons.

## Technical Skills

**Languages :** TypeScript, JavaScript (ES6+), HTML5, CSS3
**Frontend :** React.js, Vite, Tailwind CSS, Zustand, React Hook Form, GSAP, PWA
**Backend :** Node.js, Express.js, REST API Design, MVC / Modular Architecture, Socket.IO, WebSocket, BullMQ, node-cron, Zod
**Databases :** MongoDB, Mongoose, Redis (caching)
**Auth & Security :** JWT, bcrypt, RBAC, Rate Limiting, Helmet, Audit Logging, Face Verification (face-api.js)
**Integrations :** Razorpay, Nodemailer, Firebase (Auth, FCM), Cloudinary
**Tools & Deployment :** Git, GitHub, Postman, VS Code, PM2, Vercel, Render, Netlify